

GLP-1 receptor agonists and metabolic dysfunction-associated steatotic liver disease: A Systematic Review (2016-2026)

A Systematic Literature Review

Automated Literature Review Pipeline

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1 Abstract

Background: Metabolic dysfunction-associated steatotic liver disease (MASLD) is now the leading cause of chronic liver disease worldwide, affecting more than one-third of adults and tracking the parallel global epidemics of obesity and type 2 diabetes. Its progressive form, metabolic dysfunction-associated steatohepatitis (MASH), carries risk of fibrosis, cirrhosis, and hepatocellular carcinoma,

yet until the recent accelerated approvals of resmetirom and semaglutide, no pharmacotherapy was specifically approved for the disease, leaving lifestyle modification as the sole recommended intervention. Glucagon-like peptide-1 receptor agonists (GLP-1RAs) and newer dual/triple incretin-based agonists have emerged as candidate disease-modifying therapies, but the evidence base spans mechanistic, trial, and real-world comparative-effectiveness data that vary considerably by drug, disease stage, and outcome definition.

Methods: A PRISMA-informed, PubMed-only search (NCBI E-utilities) combining GLP-1RA/incretin and MASLD/MASH terminology identified 100 records; all were retained after deduplication and passed title/abstract screening, yielding 50 included studies spanning 41 journals. No risk-of-bias instrument or dual independent review was applied; single-database coverage is an acknowledged limitation.

Results: GLP-1 monotherapy, principally semaglutide, consistently reduces hepatic fat and transaminases and resolves steatohepatitis (pooled odds ratio 3.48 for MASH resolution without fibrosis worsening), but its effect on fibrosis is smaller and statistically fragile, concentrated in F2-F3 disease, with limited signal in compensated cirrhosis. Dual and triple agonists — tirzepatide (GIP/GLP-1), and survodutide, mazdutide, pemvidutide, retatrutide, and cotadutide — show larger, more consistent effects on both MASH resolution (pooled relative risk 3.32) and fibrosis improvement (relative risk 1.49), with tirzepatide achieving resolution in up to 62% of patients versus 10% with placebo. Real-world comparisons against SGLT2 inhibitors are mixed, and evidence in compensated cirrhosis and MetALD remains sparse.

Conclusion: GLP-1RA monotherapy is well-supported for steatosis-predominant, earlier-stage MASLD/MASH, while dual/triple agonism is the more defensible choice when fibrosis regression is the treatment goal. Durability after discontinuation, efficacy in advanced fibrosis and MetALD, and long-term hepatocellular carcinoma risk-benefit remain unresolved and warrant head-to-head trials and multi-database evidence synthesis.

2 Introduction

Metabolic dysfunction-associated steatotic liver disease (MASLD) has replaced non-alcoholic fatty liver disease (NAFLD) as the preferred nomenclature for hepatic steatosis occurring in the context of cardiometabolic risk, a terminology shift intended to anchor diagnosis to positive metabolic

criteria — obesity, insulin resistance, type 2 diabetes, hypertension, or dyslipidemia — rather than to the exclusion of alcohol use¹. The change is more than semantic: it is a recognition that MASLD is now the leading cause of chronic liver disease worldwide, affecting more than one-third of adults globally^{2,3}, and that its burden is inseparable from the parallel epidemics of obesity and type 2 diabetes mellitus (T2DM), which nearly tripled global obesity prevalence since 1975⁴. Within this spectrum, metabolic dysfunction-associated steatohepatitis (MASH) — the progressive, inflammatory and fibrogenic form of MASLD, historically termed non-alcoholic steatohepatitis (NASH) — represents the subset of patients at meaningful risk of cirrhosis, hepatocellular carcinoma (HCC), and liver-related mortality^{5,6}. A related but distinct entity, metabolic and alcohol-associated liver disease (MetALD), describes the overlap phenotype in which cardiometabolic risk factors coexist with concurrent alcohol exposure, producing synergistic liver injury; because most MASLD trials exclude patients with significant alcohol use, MetALD-specific evidence remains sparse and is largely extrapolated from MASLD and alcohol-associated liver disease data⁷.

Not all MASH carries equivalent prognostic weight, and the field has converged on the concept of “at-risk MASH” — steatohepatitis with fibrosis stage 2 — as the threshold that identifies patients most likely to benefit from pharmacotherapy and most likely to progress to cirrhosis if untreated⁸. Distinguishing at-risk MASH from simple steatosis matters clinically: in a prospective cohort of 413 patients undergoing liver biopsy and magnetic resonance elastography, a simplified at-risk definition captured a substantially broader group of pharmacotherapy candidates than the stricter NAFLD Activity Score-based definition (122 vs. 77 patients), and optimized liver-stiffness cutoffs on MRE (2.8–6.4 kPa) improved sensitivity for at-risk MASH from 50% to 80% relative to AASLD-recommended thresholds⁸. Getting this stratification right is central to treatment selection, since resmetirom and semaglutide — the first agents to receive accelerated approval specifically for MASH — are indicated on the basis of this at-risk designation^{6,8}. At the far end of the disease spectrum, compensated MASH-related cirrhosis remains a distinct and largely unmet therapeutic problem: concurrent metabolic dysfunction, persistent inflammation, stabilized fibrosis, and portal hypertension make this population substantially harder to treat than non-cirrhotic MASH, and although GLP-1/glucagon receptor dual agonists and fibroblast growth factor 21 analogues have advanced into trials for this stage, most candidates tested so far have not demonstrated clear histological or clinical benefit⁹.

The pathophysiological rationale for incretin-based therapy in MASLD rests on the tight mechanistic

coupling between insulin resistance, glucagon dysregulation, and hepatic lipid handling. Hepatic steatosis develops when insulin resistance drives increased adipose tissue lipolysis and de novo lipogenesis, overwhelming the liver’s capacity for fatty-acid oxidation and export, while chronic low-grade inflammation, oxidative stress, and gut microbiota alterations promote the transition from steatosis to lipotoxic injury, inflammation, and fibrogenesis¹⁰. Glucagon itself is increasingly recognized as more than a counter-regulatory hormone to insulin: it regulates energy balance and hepatic lipid metabolism, and glucagon receptor antagonism — despite potent glucose-lowering effects — has been abandoned in clinical development because it worsens hepatic steatosis and promotes weight gain, underscoring that glucagon signaling must be modulated rather than simply blocked^{11,12}. This multi-hormonal, multi-organ crosstalk is now conceptualized within the cardiovascular-kidney-liver-metabolic (CKLM), or unified cardiometabolic disease, continuum: MASLD, type 2 diabetes, atherosclerotic cardiovascular disease, heart failure with preserved ejection fraction, hypertension, and chronic kidney disease share upstream drivers — visceral adiposity, ceramide-mediated lipotoxicity, NF- κ B/NLRP3-driven inflammation, gut-derived endotoxemia, and endothelial dysfunction — such that hepatic, cardiac, and renal disease should be understood as manifestations of a single progressive process rather than independent conditions^{2,13,14}. This framing explains why cardiovascular disease, not liver failure, remains the leading cause of death in patients with MASLD³, and why therapies capable of intervening simultaneously across the CKLM axis are of particular interest. The economic and demographic scale of the problem reinforces the urgency: obesity alone accounted for an estimated 2.2% of global gross domestic product in 2019, a figure projected to exceed 3% by 2060, and its growth is now fastest in rural and low- to middle-income countries as urbanization, sedentary lifestyles, and ultra-processed food consumption spread⁴. Individual susceptibility within this metabolic milieu is not uniform: genetic variation at the PNPLA3 locus (rs738409) and at the GLP-1 receptor gene itself (GLP1R rs761387, rs6923761) has been shown to modulate treatment response, with the GLP1R rs761387 G allele independently predicting a blunted glycemic response to GLP-1RA therapy, pointing toward a future role for pharmacogenomic stratification in MASLD care¹⁵.

Glucagon-like peptide-1 receptor agonists (GLP-1RAs) were developed originally as glucose-lowering agents for T2DM, but their pleiotropic actions — appetite suppression, delayed gastric emptying, reduced hepatic lipogenesis and lipotoxicity, improved insulin sensitivity, and attenuation of systemic inflammation — have positioned them as disease-modifying therapies across the CKLM continuum^{3,16}. Liraglutide, an early GLP-1RA, demonstrated proof-of-concept efficacy in the LEAN trial, achieving

steatohepatitis resolution without fibrosis worsening in 39% of patients versus 9% with placebo, while semaglutide has shown dose-dependent histologic resolution rates up to 59% in phase 2 testing and, in an interim phase 3 analysis in F2–F3 MASH, resolution without fibrosis worsening in 62.9% of patients versus 34.3% on placebo³. Beyond mono-agonism, the field has moved toward dual and triple incretin-based agonists that co-activate the glucose-dependent insulinotropic polypeptide (GIP) receptor and, in some cases, the glucagon receptor alongside the GLP-1 receptor. Tirzepatide, a dual GIP/GLP-1 receptor agonist, achieved MASH resolution in 44–62% of patients in the SYNERGY-NASH trial (versus 10% with placebo), with concomitant fibrosis improvement in up to 51%³, and has demonstrated broader metabolic benefit across pancreatic, adipose, hepatic, cardiovascular, renal, and central nervous system targets in the SURPASS and SURMOUNT trial programs¹⁷. Triple agonists and GLP-1/glucagon receptor dual agonists such as survodutide, mazdutide, pemvidutide, retatrutide, and cotadutide are similarly under active clinical investigation, with a pooled analysis of six randomized trials (961 participants) showing that these polyagonists significantly increase MASH resolution or histological improvement without fibrosis worsening (RR 3.32, 95% CI 2.28–4.84) and fibrosis improvement without MASH worsening (RR 1.49, 95% CI 1.15–1.94) relative to placebo¹⁸. Cardiovascular outcome trials for GLP-1 monoagonists — LEADER (liraglutide), SUSTAIN-6 and SELECT (semaglutide), and REWIND (dulaglutide) — have independently established reductions in major adverse cardiovascular events, reinforcing the liver-heart axis rationale for this drug class in MASLD³.

Until the recent approvals of resmetirom and semaglutide, no pharmacotherapy had been specifically approved for MASLD or MASH, leaving lifestyle modification as the sole foundational intervention despite its well-documented difficulty to sustain⁶. This absence of approved options, combined with the rapid expansion of incretin-based agents originally developed for T2DM and obesity, has created both an opportunity and a synthesis challenge: evidence spans mechanistic studies, placebo-controlled histologic trials, real-world comparative-effectiveness analyses against sodium-glucose cotransporter-2 (SGLT2) inhibitors and other agents, and emerging data in the cardiovascular-kidney-liver-metabolic population, but findings vary considerably by drug, disease stage, comparator, and outcome definition.

Existing systematic reviews and meta-analyses of incretin-based therapy in MASLD, however, remain fragmented along drug-class and outcome-domain lines. Several are restricted to a single mono-agonist — semaglutide specifically^{19,20} or GLP-1RAs as an undifferentiated class, without

separating evidence for newer dual/triple agonists^{1,21,22} — while the syntheses that do incorporate dual and triple agonists¹⁸ or construct network meta-analyses spanning SGLT2 inhibitors and combination regimens²³ typically pool evidence from multiple proprietary databases (MEDLINE, Embase, Cochrane CENTRAL) supplemented by grey literature, limiting reproducibility for readers without institutional access, and none directly benchmarks monoagonist against dual/triple-agonist evidence within a single PRISMA-compliant framework focused specifically on MASLD/MASH liver histologic and imaging endpoints.

This gap motivates the present review: whether the incremental hepatic benefit of dual and triple incretin-based agonism over GLP-1 monotherapy is discernible and generalizable when the evidence base is restricted to a single, freely accessible, transparently reproducible source — PubMed — has not been specifically addressed. This review closes that gap by applying a PRISMA 2020-compliant, PubMed-only search strategy to systematically identify and synthesize evidence comparing GLP-1 receptor monoagonists against dual and triple incretin-based agonists on MASLD/MASH liver-related outcomes, situating these findings within the broader mechanistic, comparative-effectiveness, and safety landscape described above. This review synthesizes the current evidence on GLP-1RA and multi-agonist efficacy, safety, and clinical positioning for MASLD and MASH. Subsequent sections examine the mechanistic basis for incretin-based hepatic protection, review randomized controlled trial and meta-analytic evidence for GLP-1 monoagonists and dual/triple agonists on histologic, biochemical, and imaging endpoints, evaluate real-world and comparative-effectiveness data against SGLT2 inhibitors and other pharmacotherapies, and consider safety, patient selection, and unresolved questions — including durability of benefit, efficacy in compensated cirrhosis, and the role of these agents within the broader CKLM management framework.

This review pursues the following specific objectives:

1. To systematically identify and synthesize randomized controlled trial and observational evidence on GLP-1 receptor monoagonists (e.g., liraglutide, semaglutide, dulaglutide) for MASLD/MASH liver-related outcomes, including histologic resolution, fibrosis stage, and non-invasive imaging and biochemical markers.
2. To synthesize evidence on dual and triple incretin-based agonists (GIP/GLP-1 co-agonists, GLP-1/glucagon receptor dual agonists, and triple agonists) for the same liver-related outcomes, and to compare the magnitude and consistency of benefit against GLP-1 monotherapy.
3. To characterize the mechanistic basis for incretin-based hepatoprotection and its relationship

to the broader cardiovascular-kidney-liver-metabolic (CKLM) disease continuum.

4. To evaluate real-world and comparative-effectiveness evidence for GLP-1RAs and multi-agonists relative to SGLT2 inhibitors and other pharmacotherapies in MASLD/MASH.
5. To characterize the safety profile, tolerability, and patient-selection considerations for incretin-based therapy across disease stages, including compensated MASH-related cirrhosis.
6. To identify unresolved questions and evidence gaps — including durability of benefit, efficacy in cirrhosis, and the potential role of pharmacogenomic stratification — that should guide future research and clinical practice.

3 Methods

This systematic review was conducted following a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)-informed workflow, encompassing structured identification, screening, eligibility assessment, and inclusion of records addressing glucagon-like peptide-1 receptor agonist (GLP-1RA) therapy in metabolic dysfunction-associated steatotic liver disease (MASLD) and metabolic dysfunction-associated steatohepatitis (MASH).

3.1 Data Sources and Search Strategy

Literature was identified through a single-database search of PubMed via the National Center for Biotechnology Information (NCBI) E-utilities application programming interface. Searches of Scopus and Embase were not performed in this iteration of the review, as institutional API credentials for those databases were not configured at the time of the search; the corpus therefore reflects PubMed coverage only, and this constitutes a limitation of the present synthesis rather than a deliberate scoping decision.

The search strategy combined drug-class and agent-specific terms — “GLP-1 receptor agonist,” “incretin-based therapy,” “semaglutide,” “tirzepatide,” and related glucagon-like peptide-1-based compounds — with disease-specific terms, including “MASLD,” “MASH,” “steatotic liver disease,” “non-alcoholic fatty liver disease,” and “non-alcoholic steatohepatitis,” to ensure retrieval of records using both current (2023-onward) nomenclature and legacy terminology. Boolean operators (AND/OR) and PubMed field tags were used to combine these term sets. No language or publication-type restriction was applied at the search stage; the resulting records span the period identified during the search run.

For transparency and reproducibility, the literal query string submitted to the PubMed ESearch endpoint for this corpus, as logged by the automated search pipeline, was:

`GLP-1 receptor agonists and metabolic dysfunction-associated steatotic liver disease`

For this iteration of the review, the drug-class, agent-specific, and disease-specific term families described above were submitted as this single consolidated topic string rather than as an explicit, hand-nested Boolean expression with field tags; PubMed’s Automatic Term Mapping (ATM) expands such a string against MeSH headings and title/abstract fields to retrieve records matching its constituent concepts. No additional date restriction was appended at the query-string level.

3.2 Study Selection

Records retrieved from the initial search underwent a staged screening process. Titles and abstracts were first screened for topical relevance, followed by deduplication and a quality/eligibility pass applied to the full candidate set. Because only a single database (PubMed) was searched, no cross-database duplicate records were identified or removed at the deduplication step.

Inclusion criteria comprised: (1) human studies, including randomized controlled trials, real-world/observational studies, systematic reviews and meta-analyses, and narrative reviews or clinical guidelines; (2) studies addressing a GLP-1RA or related incretin-based therapy (including dual and triple agonists such as tirzepatide); and (3) studies reporting outcomes relevant to MASLD/MASH or closely related metabolic-hepatic endpoints (e.g., hepatic steatosis, fibrosis, liver enzymes, histologic resolution of steatohepatitis). Studies were excluded if they were purely preclinical or animal-model investigations, unless mechanistic findings were judged necessary to contextualize clinical evidence; case reports lacking generalizable methodological detail and records without an accessible abstract were also excluded. Operationally, “without an accessible abstract” was implemented as an abstract-length threshold: records were excluded during automated processing if their retrieved abstract text was shorter than 50 characters. Records were additionally excluded if they duplicated another record’s Digital Object Identifier (DOI) or, for records lacking a DOI, if their title string matched an already-retained record, consistent with the deduplication logic described above.

3.3 PRISMA Flow

Of 100 records identified through the PubMed search (identification stage), all 100 were retained following deduplication, reflecting the single-database search strategy. All 100 records passed the

subsequent quality/eligibility screen (title/abstract review against the inclusion criteria above), yielding a final analytic corpus of 50 studies included in the qualitative synthesis. The included studies were published across 41 distinct journals, indicating broad disciplinary representation spanning hepatology, endocrinology, and general internal medicine outlets.

3.4 Data Extraction and Quality Considerations

For each included record, structured metadata were extracted programmatically from PubMed, including publication year, journal, author list, abstract text, and citation indicators where available. No independent risk-of-bias or methodological-quality scoring instrument (e.g., Cochrane RoB 2, Newcastle-Ottawa Scale) was applied to individual studies in this iteration; eligibility screening functioned as the primary quality gate, and study design (RCT, observational, or review/guideline) was retained as metadata to allow downstream weighting of evidence strength during synthesis. Screening and extraction were performed using an automated pipeline rather than dual independent human reviewers; this is noted as a methodological limitation relative to full PRISMA-standard dual-reviewer protocols.

Data collection within this automated pipeline consisted of a single extraction pass rather than a multi-reviewer protocol. In addition to the bibliographic metadata above, abstract text for each record was processed with a fixed set of regular-expression patterns designed to surface quantitative data points reported in the abstract: sample sizes (e.g., “N=342” or numeric mentions of patients/participants/cohort size), statistical significance values (p-values), clinically relevant thresholds or cut-points (e.g., biomarker or laboratory values), and drug dosing information, where these were stated in the abstract text. This automated, regex-based approach substitutes for manual full-text data extraction by trained reviewers, and because a single automated pass was used rather than two or more independent extractors, there was no reconciliation or arbitration step to adjudicate extraction disagreements.

Beyond the quantitative values described above, the following variables were captured for each included study: study design/type (categorized as randomized controlled trial, real-world/observational study, systematic review/meta-analysis, or narrative review/clinical guideline), sample size (where stated), journal name, publication year, and citation count (where available from source metadata). Journal-level quality indicators — CiteScore and/or Scimago Journal Rank (SJR) quartile, where resolvable — were also recorded and are described further below.

No formal risk-of-bias or study-quality appraisal instrument was applied at the individual-study level in this pipeline: tools such as the Cochrane Risk of Bias tool (RoB 2), the Newcastle-Ottawa Scale for observational studies, or AMSTAR-2 for systematic reviews were not used to appraise the internal validity of included studies. Instead, quality control operated at the journal level, filtering candidate records against journal-level bibliometric indicators (CiteScore and/or SJR quartile) rather than against study-level bias domains. This substitution of publication-venue metrics for study-level risk-of-bias appraisal is a material limitation of the present methodology and should be borne in mind when weighing the certainty of any synthesized findings.

3.5 Data Synthesis

Given the heterogeneity of study designs (randomized controlled trials, observational studies, systematic reviews/meta-analyses, and narrative reviews/guidelines), outcome definitions, and populations represented in the included corpus, formal meta-analysis — statistical pooling of effect estimates across studies — was not undertaken. A narrative, thematic synthesis approach was used instead: no summary effect measure (e.g., pooled risk ratio, mean difference, or odds ratio) was calculated across studies, and the findings presented in this review reflect qualitative synthesis of study-level results rather than pooled quantitative estimates.

Thematic organization of the included evidence was achieved by classifying each article into one or more topic-specific categories drawn from a taxonomy generated for this review topic (spanning, for example, epidemiology/burden, pathophysiology/mechanism, GLP-1 monotherapy, dual/triple agonists, cardiometabolic comorbidities, special populations/safety, comparator/combination therapies, and clinical guidelines/definitions). This taxonomy was produced by a large language model conditioned on a sample of the retrieved titles and abstracts, rather than derived from a pre-registered, investigator-defined coding framework, and articles were assigned to categories on that basis; each body section of this review synthesizes, in narrative form, the studies assigned to its corresponding category or categories. Heterogeneity across included studies was addressed through this thematic grouping — organizing evidence by mechanism, intervention class, and clinical context — rather than through statistical heterogeneity testing (e.g., I^2 , Cochran’s Q) or subgroup meta-analysis, since no quantitative pooling was performed.

Potential publication and reporting bias in the underlying literature was not formally assessed: no funnel plot, Egger’s test, or other statistical test for small-study or publication-bias effects was

performed, consistent with the narrative (non-meta-analytic) synthesis approach used here. Likewise, no formal certainty-of-evidence framework — such as GRADE (Grading of Recommendations Assessment, Development and Evaluation) — was applied by this pipeline to rate the overall certainty of evidence for any outcome; where individual included studies reported their own GRADE assessments, these are described narratively within the relevant sections, but no independent, review-level certainty rating was generated. Both the absence of a formal publication-bias assessment and the absence of a GRADE (or equivalent) certainty rating are noted as limitations of the current methodology.

3.6 Limitations of Search Scope

Because this review draws exclusively on PubMed-indexed literature, it may under-represent conference abstracts, preprints, and journals not indexed in MEDLINE, as well as relevant records uniquely captured by Scopus or Embase. Future iterations of this review are intended to incorporate multi-database searching once the relevant API access is established.

4 Pathophysiology and Mechanistic Rationale

MASLD pathogenesis converges on insulin resistance, ectopic hepatic lipid accumulation, and chronic low-grade inflammation, which together drive lipotoxicity, hepatocyte injury, and progressive fibrogenesis toward compensated advanced chronic liver disease^{10,24}. Visceral adiposity and impaired insulin signaling promote unrestrained lipolysis and hepatic free fatty acid flux, while oxidative stress and gut microbiota alterations amplify necro-inflammatory activity within the liver¹⁰. Within this framework, glucagon has emerged as an underappreciated second axis of dysregulation alongside insulin resistance. Type 2 diabetes has traditionally been framed as a disease of insulin insufficiency, but the development of a high-accuracy glucagon sandwich enzyme-linked immunosorbent assay has clarified that hyperglucagonemia independently contributes to hyperglycemia and hepatic steatosis¹¹. Notably, glucagon receptor antagonists lower glucose effectively without hypoglycemia risk but have failed to advance past early clinical development because they paradoxically cause weight gain and worsen hepatic steatosis, whereas glucagon receptor agonism—when paired with GLP-1 receptor activation—appears to counteract this liability by increasing lipolysis, energy expenditure, and hepatic fat oxidation^{11,12}. This mechanistic reversal underlies the rationale for dual GCGR/GLP-1 receptor agonists such as survodutide and mazdutide, though their cardiac chronotropic and

blood-pressure effects require dedicated cardiovascular safety evaluation, an issue being addressed prospectively in the SYNCHRONIZE-CVOT trial¹².

The rationale for multi-receptor agonism extends beyond glucagon. Zhou and Dong propose a multi-layered conceptual framework in which GLP-1 receptor agonist efficacy in MASLD is governed not simply by pharmacological potency but by structural (hepatic and systemic signaling integrity), contextual (baseline metabolic perturbation), and temporal (durability of pharmacological input) constraints, explaining recurrent patterns of limited efficacy in advanced fibrosis, metabolically milder populations, and after treatment discontinuation¹⁶. Multi-receptor agonists that co-activate the glucose-dependent insulintropic polypeptide receptor (GIPR) and GCGR alongside GLP1R, such as tirzepatide (GIP/GLP-1) and triple agonists including retatrutide and cotadutide, are proposed to partially overcome these constraints by broadening the regulatory inputs converging on hepatocyte metabolism^{16,17}. Tirzepatide additionally increases insulin secretion, suppresses glucagon, slows gastric emptying, and shifts gut microbiota composition toward increased Bacteroidetes and decreased Firmicutes while enhancing intestinal barrier integrity, illustrating how receptor-level pharmacology intersects with a microbiota-metabolite layer of regulation¹⁷. A systematic review and meta-analysis of six randomized trials (961 participants) evaluating dual and triple GLP-1-based polyagonists—tirzepatide, survodutide, pemvidutide, retatrutide, and cotadutide—found significantly greater MASH resolution without fibrosis worsening (RR 3.32, 95% CI 2.28–4.84) and fibrosis improvement without worsening of MASH (RR 1.49, 95% CI 1.15–1.94), alongside a mean liver fat reduction of -44.60 percentage points versus placebo, supporting the premise that expanding receptor engagement yields incremental hepatic benefit over GLP-1 monoagonism¹⁸. Zhou and Dong further argue that gut microbiota and their metabolites constitute a candidate systems-level regulatory layer integrating host metabolic, immune, and endocrine signaling; convergence across diverse interventions is more consistent at the level of microbiota-associated metabolic function than taxonomic composition, though causal significance remains unproven¹⁶. This gut-pancreatic peptide axis extends further still, with glucose-dependent insulintropic polypeptide, amylin, and peptide YY under active investigation as complementary signaling nodes in the evolving mono-to-multi-agonist therapeutic landscape²⁵.

Genetic variation in lipid handling further modifies treatment response and helps explain heterogeneity in liver fat reduction. The patatin-like phospholipase domain-containing 3 (PNPLA3) I148M variant is strongly associated with steatotic liver disease, and a post hoc analysis of the SURPASS-3

MRI substudy examined whether presence of the I148M allele (genotypes GG/CG) versus its absence (genotype CC) modified the liver fat content (LFC) response to 52 weeks of tirzepatide versus insulin degludec in insulin-naïve patients with type 2 diabetes and MASLD; tirzepatide produced significant LFC, weight, and HbA1c reductions across genotype strata, indicating that the drug’s hepatic benefit is not abrogated by this common risk allele²⁶. Genetic modulation is not confined to PNPLA3: a propensity-matched genotyping study of GLP1R polymorphisms found that the rs761387 G allele independently predicted a blunted glycemic response to subcutaneous GLP-1 receptor agonist therapy ($\beta = -0.277$, $p = 0.018$), even as GLP-1RA outperformed SGLT2 inhibitors overall on HbA1c, BMI, and ALT reduction in patients with MASLD, underscoring that receptor-level pharmacogenomics, not just hepatic lipid genetics, shapes individual response¹⁵. Choline metabolism represents an adjacent, less-studied pathway in hepatic triglyceride regulation: low choline intake is linked to obesity, MASLD, and type 2 diabetes, and circulating trimethylamine N-oxide is now understood as a marker of obesity- and diet-driven metabolic dysfunction rather than a direct consequence of dietary choline; whether GLP-1 receptor agonist-induced changes in intake and body composition alter choline requirements, and thereby hepatic triglyceride handling, remains an open question warranting investigation²⁷.

Translating these mechanisms into practice requires reliable identification of the treatment-relevant population, since fibrosis stage—not simple steatosis—drives liver-related morbidity and mortality and determines eligibility for resmetirom and semaglutide^{5,6}. Inconsistent definitions of “at-risk MASH” have complicated this task. A prospective biopsy-and-MRE study of 413 patients proposed a simplified at-risk MASH definition (MASH with fibrosis stage ≥ 2) that captured a broader population ($n=122$) than the stricter NAFLD Activity Score-based definition ($n=77$), and found that MRE-derived liver stiffness (LS-MRE) alone had the highest diagnostic accuracy for at-risk MASH (AUC 0.91) and cirrhosis (AUC 0.93), outperforming FIB-4 and individual liver enzymes ($p<0.01$)⁸. Optimized LS-MRE cutoffs of 2.8–6.4 kPa, compared with AASLD-recommended cutoffs of 3.1–4.4 kPa, identified 25 additional biopsy-proven pharmacotherapy candidates—28% more eligible patients—while improving sensitivity from 50% to 80% with an acceptable trade-off in specificity (73% vs. 88%)⁸. This staging refinement has direct implications for trial design: agents with strong efficacy signals in non-cirrhotic MASH, such as fibroblast growth factor 21 analogues, have shown far less consistent benefit once fibrosis has progressed to compensated cirrhosis, reflecting disease complexity, patient heterogeneity, and the limitations of current histological endpoint frameworks in this population⁹. Together, glucagon-inclusive multi-receptor pharmacology,

microbiota-metabolite convergence, genetic modifiers of lipid and receptor response, and imaging-based fibrosis staging form the mechanistic scaffold against which the trial evidence reviewed in subsequent sections should be interpreted.

5 GLP-1 Receptor Agonist Monotherapy: Efficacy and Evidence

Prior to the emergence of incretin-based therapies, no pharmacologic agent held approval for nonalcoholic fatty liver disease (NAFLD) or its progressive form, nonalcoholic steatohepatitis (NASH), leaving weight loss through lifestyle modification as the sole recommended intervention despite its well-documented difficulty to sustain¹⁹. That landscape has shifted: semaglutide, a single-target glucagon-like peptide-1 receptor agonist (GLP-1RA), now stands alongside resmetirom as one of only two agents to have received accelerated regulatory approval for metabolic dysfunction-associated steatohepatitis (MASH), reflecting a body of placebo-controlled trial evidence that has matured enough to inform treatment guidelines⁶.

The most rigorous synthesis of this evidence comes from a systematic review and meta-analysis of placebo-controlled randomized trials of subcutaneous semaglutide in NAFLD/NASH, which applied Cochrane RoB 2 risk-of-bias assessment and GRADE certainty rating¹⁹. Semaglutide significantly reduced aspartate aminotransferase (mean difference [MD] -6.72 U/L; 95% CI -11.79 to -1.64; $P = .009$) and more than doubled the odds of NASH resolution without fibrosis progression (RR 2.14; 95% CI 1.44-3.17; $P = .0002$). Fibrosis-stage improvement, by contrast, favored semaglutide numerically but did not reach significance (RR 1.14; 95% CI 0.63-2.05; $P = .67$). Weight loss (MD -6.99%; 95% CI -13.92 to -0.06; $P = .05$) and glycated hemoglobin reduction (MD -1.29%; 95% CI -1.46 to -1.13; $P < .00001$) were the co-primary metabolic drivers underlying these hepatic effects, with gastrointestinal adverse events and discontinuation more common on active drug but serious adverse events comparable to placebo. An independent, updated meta-analysis of ten trials ($n = 1908$; PROSPERO CRD420261288562) corroborated this pattern: semaglutide markedly improved MASH resolution without worsening fibrosis (OR 3.48; 95% CI 2.68-4.53; $P < .00001$), reduced liver stiffness (MD -1.25 kPa; 95% CI -2.18 to -0.32; $P = .009$), and increased the odds of a 30% relative reduction in liver fat content (OR 7.16; 95% CI 3.08-16.64; $P < .00001$), while pooled fibrosis improvement of at least one stage again fell short of significance (OR 1.17; 95% CI 0.49-2.80; $P = .72$)²⁰. Across both syntheses, the anti-steatotic and anti-inflammatory signal is robust; the anti-fibrotic signal is directionally favorable but statistically fragile.

This histologic profile is consistent with the individual trials underpinning it. In the liraglutide LEAN trial, steatohepatitis resolution without fibrosis worsening occurred in 39% of treated patients versus 9% on placebo; semaglutide’s phase 2 dose-ranging trial showed dose-dependent resolution rates up to 59% versus 17% with placebo; and in a subsequent trial of semaglutide 2.4 mg weekly, an interim analysis in patients with F2-F3 MASH met both co-primary histologic endpoints, with resolution without fibrosis worsening in 62.9% versus 34.3% and fibrosis improvement of at least one stage in 36.8% versus 22.4% of placebo-treated patients³. A systematic review spanning twelve studies of GLP-1RA monotherapy similarly found hepatic fat reduction to be the most consistent finding across agents, demonstrated with semaglutide, liraglutide, dulaglutide, and beinaglutide, while liver-enzyme improvement was less uniform and best regarded as supportive rather than definitive; histologic benefit was strongest for steatohepatitis resolution in non-cirrhotic MASH, and fibrosis benefit clustered in F2-F3 disease with limited response once cirrhosis was established¹. A broader meta-analysis of 24 randomized trials ($n = 2158$) comparing incretin therapies with placebo, oral hypoglycemics, or insulin found consistent reductions in ALT (SMD -0.46 vs placebo; -1.02 vs insulin), AST (SMD -0.41 vs placebo; -0.62 vs insulin), and liver fat (SMD -0.62 vs insulin), with steatohepatitis resolution greater than placebo but not clearly superior to insulin, and benefits generally larger among patients with coexisting diabetes — underscoring glycemic and weight-related mechanisms as shared drivers of hepatic response²⁸.

Substantial heterogeneity nonetheless runs through this literature. An umbrella review of 23 meta-analyses applying AMSTAR 2 found that outcomes clustered into five domains — liver characteristics, enzyme levels, anthropometrics, metabolic markers, and inflammatory markers — but nearly 80% of the included reviews were of critically low methodological quality with substantial evidence overlap, leading the authors to characterize the pooled evidence base as preliminary and hypothesis-generating rather than conclusive²¹. Differences in population severity, treatment duration, and outcome definitions compound this uncertainty: a conceptual analysis proposing a multi-layered regulatory framework for GLP-1RA action specifically flags advanced fibrosis, metabolically milder populations, and treatment discontinuation as recurring settings of limited therapeutic response, suggesting that indirect, systemic mechanisms of hepatic benefit may saturate once intrahepatic or systemic signaling capacity is exhausted¹⁶. Even the definition of the population meant to benefit is unsettled; work applying magnetic resonance elastography (MRE) in 413 biopsied patients showed that a simplified at-risk MASH definition (fibrosis stage 2) captured substantially more treatment candidates than a stricter NAFLD Activity Score-based definition (122 vs 77 patients), and that

optimized liver-stiffness cutoffs (2.8-6.4 kPa) identified 28% more biopsy-proven pharmacotherapy candidates than AASLD-recommended thresholds, improving sensitivity from 50% to 80% — a reminder that eligibility criteria alone can materially shift apparent trial effect sizes⁸.

The advanced fibrosis and compensated cirrhosis population remains the starkest evidence gap. Few placebo-controlled trials enroll patients with F4 compensated cirrhosis (F4c), but a phase 2, dose-ranging trial that included this population is instructive: among 698 randomized participants, semaglutide 2.4 mg monotherapy achieved fibrosis improvement with no MASH worsening at week 52 in 30% of patients versus 16% on placebo, a nominally significant difference (estimated difference in responder proportions 14.05; 95% CI 1.88-26.23; $P = .024$), even though combining semaglutide with the investigational agent zalfermin did not significantly outperform placebo (24% vs 16%; $P = .19$)²⁹. This trial offers rare direct evidence that semaglutide monotherapy retains an anti-fibrotic signal even in a cirrhosis-inclusive population, though the effect size is modest and the trial was not powered for this comparison alone.

Real-world comparative evidence against sodium-glucose cotransporter-2 inhibitors (SGLT2i) is similarly mixed rather than uniformly favorable to GLP-1RAs. A propensity-matched TriNetX cohort of over 180,000 patients found semaglutide initiators had markedly lower one-year hazards of mortality (HR 0.38), hospitalization (HR 0.44), major cardiovascular events (HR 0.50), and major adverse liver outcomes (HR 0.36) compared with SGLT2i initiators, with attenuated but still favorable hazard ratios at five years³⁰, and a genotype-stratified propensity-matched study similarly found subcutaneous GLP-1RA superior to SGLT2i for HbA1c and BMI reduction and the only class producing significant ALT reduction in the MASLD subgroup ($P = .02$)¹⁵. Yet a separate propensity-matched cohort of 705 patients with type 2 diabetes and NAFLD reported the opposite direction for liver enzymes, with SGLT2i producing significantly greater ALT and AST reductions than GLP-1RAs at six months ($P = .011$ and $P = .010$, respectively)³¹, and an exploratory meta-analysis of real-world comparative cohorts versus DPP-4 inhibitors found only a modest, very-low-certainty signal for reduced incident cirrhosis or composite serious liver events (HR 0.85; 95% CI 0.74-0.98)²². Beyond the liver-specific comparisons, liraglutide 3.0 mg initiated early after bariatric surgery produced dose-dependent total weight loss (16.3% at six months vs 5.3% with standard care; $P < .05$) alongside remission of MASLD and hyperuricemia, reinforcing weight loss magnitude as a proximate mediator of hepatic benefit independent of the specific comparator³².

Taken together, the totality of placebo-controlled and comparative evidence supports GLP-1RA

monotherapy, and semaglutide specifically, as an effective intervention for steatosis regression, biochemical improvement, and steatohepatitis resolution in non-cirrhotic MASLD/MASH, with weight loss and glycemic control as the principal mediating drivers. The fibrosis and advanced-disease evidence base is comparatively thin: pooled fibrosis-stage improvement remains statistically inconsistent, GRADE-rated certainty is generally low to very low for hard liver outcomes, and only limited trial data — largely from a single phase 2 study — directly address compensated cirrhosis. This asymmetry between strong steatohepatitis-resolution evidence and fragile fibrosis evidence is precisely what current guideline updates must reconcile as GLP-1RAs move from adjunctive metabolic therapy toward a foundational role in MASLD pharmacotherapy.

6 Dual and Triple Agonists: Tirzepatide and Emerging Polyagonists

Single-target GLP-1 receptor agonism plateaus in a recognizable pattern: benefit is most reliable for steatosis and inflammation, more inconsistent for advanced fibrosis, and prone to erosion after treatment withdrawal. Zhou and Dong¹⁶ frame this as a structural limitation rather than simply insufficient potency — GLP-1R-based therapies act predominantly through indirect, systemic metabolic regulation, so their efficacy is bounded by the integrity of intrahepatic and systemic signaling networks, the severity of baseline metabolic derangement, and the persistence of drug exposure. Multi-receptor agonists are positioned as a direct response to this ceiling, expanding regulatory input by co-activating the glucose-dependent insulinotropic polypeptide receptor (GIPR) and, in newer molecules, the glucagon receptor (GCGR) alongside GLP-1R.

Tirzepatide, the first approved dual GIP/GLP-1 receptor agonist, illustrates this expanded mechanism in clinical practice. As a multi-organ integrator, it raises insulin secretion, suppresses glucagon, slows gastric emptying, and produces weight loss substantially greater than GLP-1 monoagonism, with downstream effects on adipose tissue, gut microbiota composition, and intestinal barrier integrity that plausibly reinforce its hepatic actions¹⁷. The clearest imaging evidence for a hepatic effect independent of glycemic control comes from the SURPASS-3 MRI substudy, in which tirzepatide significantly reduced MRI-assessed liver fat content compared with insulin degludec in insulin-naïve patients with type 2 diabetes and MASLD²⁶. A post hoc analysis of that substudy tested whether this benefit depended on genetic steatosis risk, comparing liver fat and cardiometabolic changes at 52 weeks by presence (GG/CG) or absence (CC) of the PNPLA3 I148M variant; tirzepatide-treated

participants showed significant reductions in liver fat content, weight, and HbA1c regardless of genotype, suggesting the drug’s hepatic effect is not contingent on a specific lipogenic risk allele²⁶. The SYNERGY-NASH trial extends this signal to histology, with tirzepatide achieving MASH resolution in 44%-62% of patients versus 10% on placebo and fibrosis improvement in up to 51% versus 30%³.

Real-world data reinforce the trial findings while adding mechanistic texture. In a retrospective multicenter study of 23 adults with obesity and MASLD (fatty liver index [FLI] > 44) treated with tirzepatide 2.5-5 mg weekly for three months, body weight, BMI, waist circumference, fasting glucose, HbA1c, HOMA-IR, hs-CRP, liver enzymes, and FLI all declined significantly ($p < 0.001$ for most comparisons)³³. Change in FLI correlated with change in weight, BMI, waist circumference, hs-CRP, HbA1c, AST, and HDL cholesterol, but on multiple regression only Δ C-reactive protein remained an independent predictor of Δ FLI, explaining 29.8% of the variance — implicating reduced low-grade inflammation, not weight loss alone, as a driver of hepatic improvement³³. A complementary question — whether switching from an existing GLP-1RA to tirzepatide adds incremental hepatic benefit — was addressed by the Hokkaido-TZP study, a retrospective secondary analysis of 182 patients with type 2 diabetes at high MASLD risk who converted from GLP-1RA therapy to tirzepatide³⁴. Over six months, switching reduced HbA1c, body weight, and liver enzymes, and lowered the Hepatic Steatosis Index in both fibrosis-risk strata; critically, a significant reduction in FIB-4 index emerged only in the high fibrosis-risk group ($P < 0.001$), and this improvement was not correlated with change in BMI or HbA1c — baseline HSI instead correlated with the magnitude of FIB-4 change ($r = 0.234$, $P = 0.005$)³⁴. Together these two studies suggest tirzepatide’s fibrosis-related benefit is partly dissociated from its weight and glycemic effects, consistent with a genuinely broader receptor-level mechanism rather than a more potent version of the same GLP-1 pathway.

Beyond tirzepatide, a growing evidence base now speaks to dual and triple GLP-1-based polyagonists as a class. A systematic review and meta-analysis pooling six randomized trials (961 participants) of tirzepatide, survodutide, pemvidutide, retatrutide, and cotadutide against placebo found that polyagonist therapy significantly increased MASH resolution or histological improvement without fibrosis worsening (RR 3.32, 95% CI 2.28-4.84; $I^2 = 20\%$) and fibrosis improvement without MASH worsening (RR 1.49, 95% CI 1.15-1.94; $I^2 = 0\%$), alongside a much larger relative reduction in liver fat content than has typically been reported for GLP-1 monotherapy (MD -44.60 percentage points, 95% CI -51.17 to -38.04) and a markedly increased likelihood of achieving at least a 30%

relative reduction in MRI-PDFF (RR 4.71, 95% CI 3.04-7.30)¹⁸. Gastrointestinal adverse events (nausea, diarrhea, vomiting) were more frequent with polyagonists, though serious adverse events and treatment discontinuation did not differ significantly from placebo¹⁸. A network meta-analysis incorporating these newer agents alongside SGLT2 inhibitors and FGF21 analogs across 34 studies ranked GLP-1/GIP dual agonists highest of all drug classes for MASH resolution (RR 5.21), ahead of FGF21 analogs (RR 3.52) and SGLT2 inhibitors (RR 2.92); FGF21 analogs and SGLT2 inhibitors instead led on fibrosis improvement specifically (RR 2.22 and RR 2.27, respectively), with GLP-1 monoagonists trailing (RR 1.51)²³. This dissociation — polyagonists excelling at inflammatory/histological resolution, other mechanisms edging ahead on fibrosis regression — underscores that “beyond GLP-1” does not yet mean a single agent dominates every histological endpoint.

Glucagon receptor co-agonism introduces a distinct cardiovascular consideration not present with GIP co-agonism. Because glucagon regulates energy expenditure and lipid metabolism in addition to glucose homeostasis, GCGR/GLP-1 dual agonists such as mazdutide and survodutide were designed to extend metabolic benefit, but exogenous glucagon also has acute chronotropic, inotropic, and hypertensive effects, raising a theoretical safety question even though unequivocal evidence of GCGR expression in human myocardium is lacking¹². In clinical studies, mazdutide and survodutide increased heart rate to an extent similar to GLP-1 monoagonists — a class already considered cardioprotective despite this chronotropic effect — while also lowering blood pressure and improving hyperglycemia; however, at least one other GCGR/GLP-1 dual agonist was discontinued in development partly because of unacceptably large heart-rate increases and QT-interval prolongation, a difference the authors attribute to varying relative potency at the two receptors¹². Dedicated cardiovascular outcomes data are still pending, notably the SYNCHRONIZE-CVOT trial evaluating survodutide and a companion thorough QT study, meaning glucagon receptor co-agonism carries real, compound-specific cardiovascular uncertainty that GIP co-agonism does not¹².

Taken together, this evidence supports dual and, increasingly, triple agonism as the mechanistic strategy most likely to close the gap between MASLD’s inflammatory and fibrotic components — with dual GLP-1/GIP agonism already showing superior metabolic control and steatosis reduction, even as definitive histological confirmation of fibrosis reversal remains incomplete for several agents in the class¹⁰. Broader incretin-class meta-analyses (24 RCTs, n = 2158) confirm consistent reductions in ALT, AST, and liver fat relative to placebo and insulin, but note that effects on histologic fibrosis

remain inconclusive across the incretin category as a whole, reinforcing that fibrosis resolution — not steatosis or inflammation — is where the efficacy ceiling of GLP-1-based therapy, single or multi-receptor, is still most evident²⁸. Whether multi-receptor agonism can also address the hardest population — compensated MASH-related cirrhosis — remains unresolved: GLP-1R/GCGR dual agonists are among the agents now being tested in this setting, but the predominance of negative trials for cirrhosis-stage disease suggests that receptor expansion alone may not overcome the added burden of stabilized fibrosis, portal hypertension, and patient heterogeneity that characterizes advanced disease⁹.

7 Cardiometabolic Comorbidities and Special Clinical Contexts

MASLD is increasingly framed not as an isolated hepatic diagnosis but as one node within a broader cardiovascular-kidney-liver-metabolic (CKLM) or cardiovascular-kidney-metabolic (CKM) syndrome that also encompasses type 2 diabetes, atherosclerotic cardiovascular disease (ASCVD), heart failure with preserved ejection fraction (HFpEF), hypertension, and chronic kidney disease (CKD)¹³. A structured narrative synthesis proposes a “unified cardiometabolic disease” continuum in which visceral adipose tissue dysfunction, ceramide-mediated lipotoxicity, NF- κ B/NLRP3-driven inflammation, gut-derived endotoxemia and TMAO, adipokine dysregulation, and a cardiorenal amplification loop drive progression across all of these phenotypes, with stage-specific mediators serving as candidate biomarkers¹³. This framing is reinforced by epidemiologic data showing that MASLD, now affecting more than one-third of adults worldwide, coexists with CKD and cardiovascular disease and independently predicts kidney function decline, cardiovascular events, and mortality, yet remains underrecognized in nephrology and cardiology practice². A parallel “dual-heart-liver” model highlights bidirectional epidemiologic and mechanistic links between MASLD and coronary artery disease—endothelial dysfunction, chronic inflammation, insulin resistance, and shared genetic susceptibility—and calls for systematic cardiovascular risk assessment in every MASLD patient³⁵. Obesity-driven complications extend further still, encompassing obstructive sleep apnea, dyslipidemia, and obesity-related glomerulopathy mediated by renin-angiotensin-aldosterone activation and glomerular hyperfiltration³⁶, against a backdrop of a global obesity epidemic that has nearly tripled in prevalence since 1975 and now underpins cardiovascular, hepatic, renal, and oncologic disease burden worldwide⁴. Traditional and emerging predictors of incident ASCVD in patients with type 2 diabetes and MASLD are being reassessed within this same continuum³⁷,

and state-of-the-art reviews increasingly frame MASLD pharmacotherapy selection around this complication-centric, multisystem model rather than liver-only endpoints^{5,6}.

Within this CKM framework, GLP-1 receptor agonists and tirzepatide are increasingly positioned against or alongside other cardiometabolic agents. Fibrates, reappraised through the CKLM lens, activate PPAR- and show the strongest evidence for slowing diabetic retinopathy progression, with post hoc signals for reduced albuminuria/eGFR decline and only modest hepatic benefit from fenofibrate—positioning fibrates as a complement to, not a substitute for, incretin-based therapy for residual microvascular risk¹⁴. Head-to-head comparisons with SGLT2 inhibitors are accumulating rapidly. In a propensity-matched retrospective study, subcutaneous GLP-1RAs produced superior HbA1c and BMI reduction versus SGLT2 inhibitors ($\beta = -0.349$, $p = 0.001$; $\beta = -0.711$, $p = 0.045$) and were the only class associated with significant ALT reduction in patients with elevated baseline ALT or hepatic steatosis ($p = 0.02$), with the GLP1R rs761387 variant identified as an independent predictor of blunted glycemic response¹⁵. A large TriNetX-based propensity-matched cohort comparing semaglutide ($n = 128,332$) with SGLT2 inhibitors ($n = 100,542$) found semaglutide associated with markedly lower 1-year hazards of mortality (HR 0.382), hospitalization (HR 0.444), MACE (HR 0.502), and major adverse liver outcomes (HR 0.361), with attenuated but persistent benefit at 5 years³⁰. A German claims-based cohort of 45,256 people with MASLD and type 2 diabetes similarly found GLP-1RA use associated with a lower hazard of major adverse liver outcomes versus SGLT2 inhibitors (HR 0.91, strengthening to HR 0.78 on-treatment), including reduced cirrhosis (HR 0.88) and hepatocellular carcinoma (HR 0.76), though without a difference in all-cause mortality³⁸. Target trial emulations using TriNetX further suggest GLP-1RA plus SGLT2i combination therapy reduces mortality (corrected HR 0.655) and composite liver outcomes (corrected HR 0.907) versus SGLT2i monotherapy, but carries increased MACE risk versus GLP-1RA monotherapy alone (corrected HR 1.111)—a trade-off relevant to sequencing³⁹. A network meta-analysis of 34 histologic MASH trials found SGLT2 inhibitors numerically outperforming GLP-1RAs for fibrosis improvement (RR 2.27 vs RR 1.51), with FGF21 analogues (RR 2.22) and GLP-1/GIP dual agonists (RR 5.21 for resolution) ranking highest overall—underscoring that GLP-1RA monotherapy is not the top-performing agent on every histologic endpoint even as it remains foundational for cardiometabolic risk reduction^{1,23}.

A distinct evidentiary gap concerns metabolic and alcohol-associated liver disease (MetALD), an emerging phenotype combining cardiometabolic risk factors with concurrent alcohol exposure that

produces synergistic liver injury and faster fibrosis progression. Because most MASLD trials exclude patients with ongoing alcohol use and alcohol-associated liver disease trials focus on severe alcoholic hepatitis, dedicated MetALD evidence is essentially absent, forcing clinicians to extrapolate from both literatures. Current management integrates alcohol-use control—prioritized as the dominant driver given its faster disease trajectory, using acamprosate or naltrexone alongside psychosocial intervention—with cardiometabolic therapies including incretin-based agents, FGF21 analogues, PPAR agonists, and thyroid hormone receptor- agonists in selected patients, plus longitudinal noninvasive fibrosis monitoring given the dynamic nature of both exposures⁷.

Special clinical contexts further shape GLP-1RA positioning. In inflammatory bowel disease, more than half of patients with Crohn’s disease or ulcerative colitis are overweight or obese at diagnosis, and visceral adipose tissue—independent of BMI—drives disease activity, reduced therapeutic response, and surgical risk while amplifying comorbid MASLD and cardiovascular disease. Preclinical data show GLP-1RAs reduce visceral adiposity, modulate gut microbiota, and attenuate proinflammatory signaling relevant to IBD pathogenesis, and emerging human data suggest GLP-1RA therapy is safe in IBD and associated with lower hospitalization and surgical risk, though effects on luminal disease activity remain unclear, warranting multidisciplinary care attentive to GI tolerability and lean mass loss⁴⁰. Gastroenterologists are also increasingly central to peri-procedural GLP-1RA management: because these agents slow gastric emptying, they raise the incidence of retained gastric contents and procedural disruption at endoscopy, yet real-world data show no increase in aspiration pneumonia among GLP-1RA users undergoing colonoscopy (approximately 0.01%-0.05%) and only a small absolute rise in aspiration-related events during upper endoscopy (fewer than 0.3% of procedures). Contemporary guidance therefore favors continuing GLP-1RA therapy with structured risk assessment, selective point-of-care gastric ultrasound, and a pre-procedural clear liquid diet rather than routine cessation⁴¹, consistent with gastroenterologists’ expanding role across obesity medicine, GLP-1 therapy, and metabolic care⁴².

Cirrhosis management in patients with diabetes presents distinct pharmacologic constraints. As MASLD becomes the dominant driver of cirrhosis, hepatogenous diabetes complicates diagnosis (HbA1c loses reliability in advanced liver disease) and treatment: metformin remains first-line in compensated cirrhosis, GLP-1RAs and SGLT2 inhibitors show promise specifically in compensated disease with cardiorenal comorbidity, sulfonylureas and fluid-retaining agents require caution as hepatic reserve declines, and insulin remains the cornerstone once decompensation or hospitalization

occurs, with conservative titration to avoid hypoglycemia⁴³. National guidelines echo this stepwise approach, recommending GLP-1RA consideration in eligible patients with type 2 diabetes or obesity while reserving multidisciplinary management, HCC surveillance, and transplant evaluation for decompensated cirrhosis^{10,44}. Trial evidence in compensated cirrhosis remains disappointing overall: a phase 2 trial testing zalfermin combined with semaglutide in participants with F2-F4c fibrosis (including compensated cirrhosis) found the combination did not significantly improve fibrosis without MASH worsening versus placebo (24% vs 16%; $p = 0.19$), though semaglutide alone showed a nominally significant effect (30% vs 16%; $p = 0.024$), suggesting semaglutide alone—rather than combination strategies—merits further evaluation in this population²⁹. This mirrors a broader pattern across MASH-cirrhosis drug development in which most investigational agents have failed to show clear histologic benefit, with FGF21 analogues currently the most promising class and GLP-1/glucagon dual agonists among several approaches still advancing⁹.

GLP-1RAs also carry a dual-edged relationship with hepatocellular carcinoma (HCC) risk. Real-world studies and target trial emulations suggest GLP-1RA use is associated with lower rates of hepatic decompensation, cirrhosis progression, and incident HCC, with the strongest apparent benefit observed against insulin or sulfonylureas and attenuated differences versus other metabolically favorable comparators such as SGLT2 inhibitors—consistent with the HCC hazard ratio of 0.76 reported in the German MASLD-diabetes cohort—though effects in established cirrhosis appear less consistent and causality remains unproven^{38,45}. Finally, as multi-receptor agonism advances toward glucagon receptor co-activation, cardiovascular safety within this CKM population warrants scrutiny: GCGR/GLP-1 dual agonists such as mazdutide and survodutide raise heart rate to a degree similar to cardioprotective GLP-1 monoagonists while also lowering blood pressure and glycemia, but at least one dual agonist was discontinued for excessive heart-rate increases and QTc prolongation, and compound-specific effects are being clarified in dedicated outcomes trials including SYNCHRONIZE-CVOT¹².

8 Discussion and Future Directions

Across the corpus synthesized in this review, a consistent hierarchy emerges. GLP-1 receptor agonist monotherapy — chiefly semaglutide — reliably resolves steatohepatitis and reduces liver fat and transaminases, but its effect on fibrosis is smaller, less consistent, and stage-dependent. An updated meta-analysis of ten trials ($n = 1,908$) found semaglutide significantly improved MASH resolution

without worsening fibrosis (OR 3.48, 95% CI 2.68-4.53) while pooled fibrosis improvement fell short of significance (OR 1.17, 95% CI 0.49-2.80), with histological reversal concentrated in non-cirrhotic disease²⁰. A GRADE-assessed systematic review of placebo-controlled trials¹⁹ and a twelve-study systematic review spanning semaglutide, liraglutide, dulaglutide, and beinaglutide¹ reach the same conclusion: hepatic fat and biochemical markers respond consistently, fibrosis regression is real but mainly confined to F2-F3 disease, and an umbrella review of 23 meta-analyses found broad benefit across liver, anthropometric, and metabolic domains but rated nearly 80% of the underlying reviews as critically low quality by AMSTAR 2²¹. Dual and triple agonism narrows this fibrosis gap. A meta-analysis of six trials (961 participants) of tirzepatide, survodutide, pemvidutide, retatrutide, and cotadutide showed a markedly larger effect on both MASH resolution (RR 3.32) and fibrosis improvement (RR 1.49) than has been reported for GLP-1 monotherapy¹⁸, and a 34-study network meta-analysis ranked GLP-1/GIP dual agonism highest for MASH resolution (RR 5.21)²³. Tirzepatide’s SURPASS-3 MRI substudy and its PNPLA3-stratified post hoc analysis showed liver fat reduction independent of genotype²⁶, and SYNERGY-NASH achieved MASH resolution in 44-62% of patients versus 10% on placebo³. The clinical takeaway is that monotherapy remains adequate — and better studied — for steatosis-predominant, earlier-stage disease, while multi-receptor agonism is the more defensible choice when fibrosis is the treatment target, consistent with the structural framework proposed by Zhou and Dong, in which GLP-1R action is bounded by systemic signaling integrity, baseline metabolic severity, and persistence of exposure¹⁶.

Read against the broader MASLD and metabolic-disease treatment literature, these findings reflect a decisive shift away from lifestyle intervention as the sole first-line strategy and toward structured pharmacologic management, paralleling an analogous shift underway in obesity medicine more generally⁴. Contemporary reviews of the MASLD therapeutic landscape increasingly frame GLP-1-based agents — alongside resmetirom and emerging multi-agonists — as core components of a stepwise, biomarker-guided treatment algorithm rather than as adjuncts to weight-loss counseling, reflecting growing recognition that diet and exercise alone rarely achieve or sustain the magnitude of weight loss needed for histological regression in most patients with MASH⁶. This reframing extends beyond hepatology: gastroenterologists and other non-endocrinology specialists are increasingly positioned as primary prescribers of GLP-1 therapy for metabolic disease, reflecting its expanding role across the obesity-MASLD-cardiometabolic continuum rather than a liver-specific niche⁴². The mono-to-multi-agonist trajectory documented in this review for hepatic endpoints mirrors a parallel evolution across gut-pancreatic peptide therapeutics more broadly, in which increasing

receptor multiplicity is being pursued as a general strategy to close the efficacy gap left by lifestyle intervention and single-target pharmacotherapy alike²⁵. Because MASLD rarely occurs in isolation from cardiovascular, renal, and broader cardiometabolic disease, the positioning of GLP-1-based therapy set out in this review is best understood as one component of an integrated cardiometabolic treatment strategy rather than a liver-disease-specific intervention^{2,13}.

Comparative-effectiveness data against SGLT2 inhibitors reinforce this positioning rather than displacing it. In a propensity-matched real-world cohort, subcutaneous GLP-1RAs outperformed SGLT2 inhibitors on HbA1c, BMI, and ALT reduction¹⁵, and initiating semaglutide rather than an SGLT2 inhibitor was associated with markedly lower five-year hazards of mortality, hospitalization, major adverse cardiovascular events, and major adverse liver outcomes in a 55,525-patient matched TriNetX cohort³⁰. A German claims-based cohort similarly found GLP-1RA use associated with lower hazards of major adverse liver outcomes and cirrhosis than SGLT2 inhibitors (HR 0.91 and 0.88, respectively)³⁸. Yet combining the two classes is not uniformly superior: three pairwise target trial emulations found GLP-1RA-plus-SGLT2i combination therapy reduced mortality and liver outcomes versus SGLT2i monotherapy, but increased MACE risk relative to GLP-1RA monotherapy alone (corrected HR 1.111)³⁹ — a signal that argues against reflexive combination therapy and for outcome-specific sequencing decisions.

Guideline integration is beginning to formalize this evidence, though unevenly across drug classes and disease stages. The first national MASLD guideline identified in this corpus, from Bosnia and Herzegovina, embeds GLP-1RAs within a stepwise, primary-care-anchored care pathway: FIB-4 as first-line risk stratification, elastography for indeterminate or high-risk results, and GLP-1RA consideration in eligible patients with type 2 diabetes or obesity, reserving bariatric surgery and multidisciplinary cirrhosis management for advanced disease⁴⁴. This stepwise logic depends on accurate at-risk MASH identification, which remains inconsistently defined across the literature. A prospective biopsy-and-MRE study of 413 patients proposed a simplified at-risk MASH definition (fibrosis stage 2) and optimized liver-stiffness cutoffs on magnetic resonance elastography (2.8-6.4 kPa) that captured 28% more biopsy-confirmed pharmacotherapy candidates than AASLD-recommended thresholds, improving sensitivity from 50% to 80% for identifying patients eligible for resmetirom or semaglutide⁸. Genetic stratification adds a further layer: PNPLA3 I148M genotype did not modify tirzepatide’s hepatic fat response²⁶, but a GLP1R variant (rs761387) was independently associated with a blunted glycemic response to GLP-1RA therapy¹⁵, suggesting that

receptor-level pharmacogenomics — distinct from lipogenic risk genotype — may eventually inform agent selection even as it is not yet actionable in routine practice.

Several questions remain genuinely unresolved. Durability after discontinuation is the most consequential: because GLP-1R-based hepatic benefit operates substantially through indirect, weight- and inflammation-mediated pathways rather than direct antifibrotic action, benefit is plausibly contingent on continued drug exposure, and none of the trials in this corpus reported long-term outcomes after treatment withdrawal¹⁶. Applicability to advanced fibrosis, cirrhosis, and MetALD is similarly unsettled. The phase 2 zalfermin-semaglutide trial in patients with F2-F4c fibrosis, including compensated cirrhosis, found that the zalfermin-semaglutide combination did not significantly improve fibrosis over placebo, though semaglutide alone showed a nominally significant effect (30% vs 16% responders, $p = 0.024$) — evidence that semaglutide retains some activity in compensated cirrhosis even as combination strategies have not yet cleared this bar²⁹. More broadly, compensated MASH-related cirrhosis is characterized by a predominance of negative trials across drug classes, reflecting disease complexity and endpoint limitations as much as inadequate drug potency⁹, and diabetic cirrhosis management guidance still restricts confident GLP-1RA use to compensated disease⁴³. MetALD — alcohol-associated injury superimposed on cardiometabolic risk — has no dedicated GLP-1RA trials at all; current practice extrapolates from separate MASLD and alcohol-associated liver disease evidence bases, an acknowledged gap rather than a validated extrapolation⁷.

Safety in special procedural and inflammatory-disease populations is better characterized but still evolving. GLP-1RA-associated delayed gastric emptying increases retained gastric contents and procedural disruption during endoscopy, yet real-world aspiration pneumonia rates remain low (approximately 0.01-0.05% during colonoscopy, under 0.3% during upper endoscopy), supporting current guidance to continue therapy with structured dietary and procedural precautions rather than routine cessation⁴¹. In inflammatory bowel disease, GLP-1RAs appear safe and are associated with lower hospitalization and surgical risk, though effects on IBD disease activity itself remain unclear pending ongoing randomized trials⁴⁰. The long-term hepatocellular carcinoma risk-benefit balance is the least settled question of all: real-world studies and target trial emulations suggest lower HCC incidence with GLP-1RA use, particularly against insulin or sulfonylureas, but this advantage attenuates in comparisons with other metabolically favorable agents such as SGLT2 inhibitors⁴⁵, a pattern echoed by the nonsignificant HCC hazard reduction (HR 0.76) in the German

cohort³⁸; causal inference here is limited by confounding by indication and by the fact that no trial in this corpus was powered for HCC as a primary endpoint.

This review’s own evidence base carries limitations that temper these conclusions. The literature is heterogeneous in outcome definition — histological fibrosis stage, MRI-PDFF, FIB-4, and liver stiffness are used interchangeably as “hepatic benefit” across studies — and surrogate and imaging endpoints substantially outnumber hard histologic or cardiovascular outcome trials; an exploratory meta-analysis of real-world cohorts rated its own pooled estimate for advanced liver outcomes as very-low-certainty evidence, appropriately describing it as hypothesis-generating rather than definitive²². The review process itself adds further constraints: literature was drawn exclusively from PubMed, without Scopus or Embase coverage, so conference abstracts, preprints, and journals outside MEDLINE indexing are underrepresented; screening and extraction used an automated single-pass pipeline rather than dual independent human review or a formal risk-of-bias instrument; no quantitative meta-analysis was performed, limiting this synthesis to qualitative pattern description; potential publication and reporting bias within the underlying primary and secondary literature — for instance, selective reporting or publication of favorable MASH-resolution or fibrosis-improvement outcomes — was not formally assessed, and no funnel plot or other quantitative test for small-study effects was performed, a limitation inherent to this review’s narrative rather than meta-analytic design; and only English-language, abstract-accessible records were retrievable through the search pipeline, introducing a plausible language and publication bias against non-English regional guidelines and trials.

Future research priorities follow directly from these gaps. Head-to-head trials comparing polyagonists against GLP-1RA monotherapy — rather than each against placebo separately — are needed to establish whether the fibrosis advantage seen for dual/triple agonists in indirect comparisons survives direct testing. Biomarker-guided patient selection, combining MRE-based at-risk MASH thresholds⁸ with emerging pharmacogenomic signals^{15,26}, should be prospectively validated as an enrollment and treatment-selection strategy rather than used only retrospectively. Given that MASLD increasingly sits within a broader cardiovascular-kidney-liver-metabolic syndrome framework^{2,13}, outcomes research should extend beyond liver-specific endpoints to integrated multi-organ trial designs, and dedicated fibrate and combination-therapy reappraisals suggest complementary, non-incretin pathways still warrant testing alongside GLP-1-based regimens¹⁴.

In sum, the evidence assembled here supports a two-tier clinical approach: GLP-1 receptor agonist

monotherapy, principally semaglutide, as an effective and well-characterized option for steatosis-predominant MASLD and earlier-stage MASH, and dual or triple agonism — led by tirzepatide, with polyagonists such as survodutide, pemvidutide, retatrutide, and cotadutide close behind — reserved for patients where fibrosis regression, not just steatohepatitis resolution, is the treatment goal⁶. National guideline frameworks are beginning to operationalize this logic through stepwise, biomarker-anchored care pathways, but major uncertainties persist around durability after discontinuation, efficacy in compensated cirrhosis and MetALD, and the long-term hepatocellular carcinoma risk-benefit balance. These gaps, together with this review’s PubMed-only, single-pass, non-meta-analytic methodology, mean the present synthesis should be read as an organizing map of a rapidly moving field rather than a definitive clinical practice statement — one that will need systematic updating as head-to-head polyagonist trials, cirrhosis-stage outcome data, and multi-database evidence syntheses mature.

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